

# Graphs and characteristics of quadratic functions



1

- a Maximum
- b 25

2

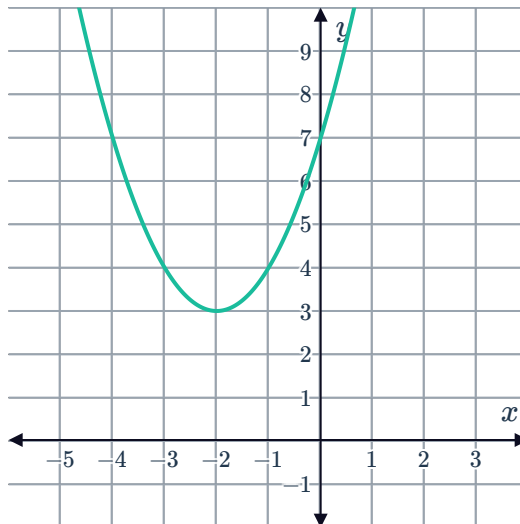
- a  $x = -10$
- b  $m = -697$

3

- a The function has a minimum value.
- b 4
- c  $-16$
- d  $(4, \infty)$
- e  $(-\infty, 4)$

4

a



- b The function has a minimum value.
- c 3
- d  $(-2, \infty)$
- e  $(-\infty, -2)$

5

- a No

c No

d Yes



6

a No

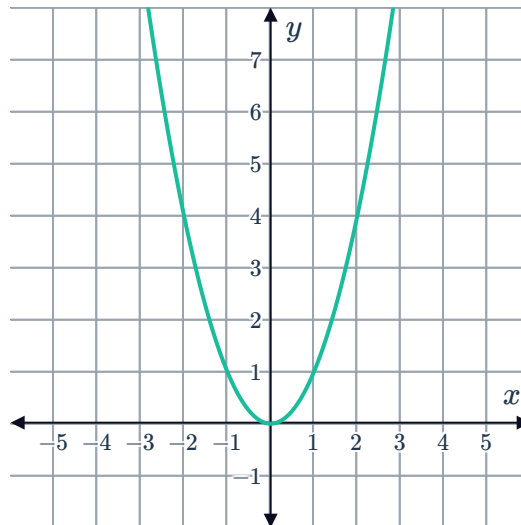
b  $-\frac{b}{2a}$

7

a

|     |    |    |    |   |   |   |   |
|-----|----|----|----|---|---|---|---|
| $x$ | -3 | -2 | -1 | 0 | 1 | 2 | 3 |
| $y$ | 9  | 4  | 1  | 0 | 1 | 4 | 9 |

b



c No

d  $x = 0$

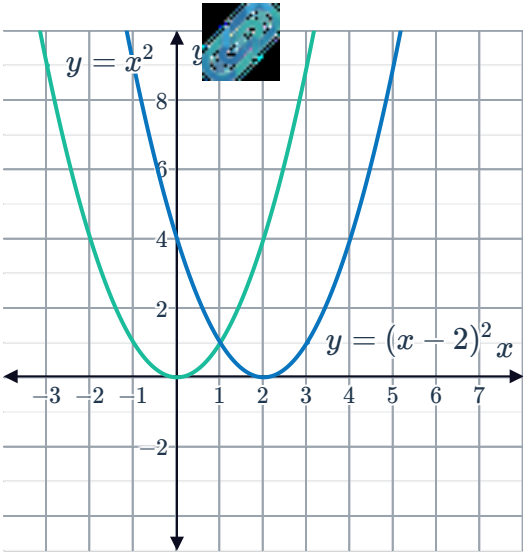
e 0

f 2

8

a Move the graph to the right by 2 units.

b

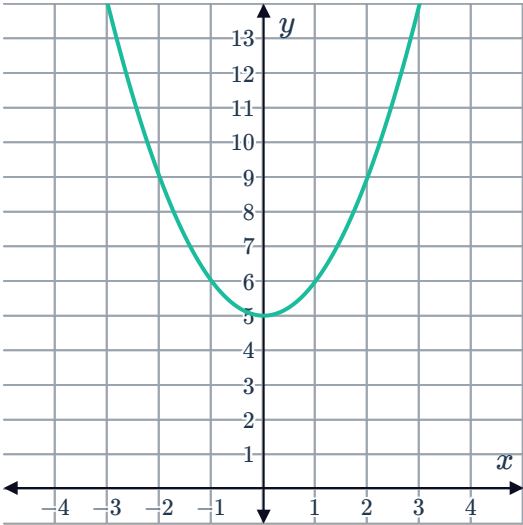


9

a

|     |    |    |    |   |   |   |    |
|-----|----|----|----|---|---|---|----|
| $x$ | -3 | -2 | -1 | 0 | 1 | 2 | 3  |
| $y$ | 14 | 9  | 6  | 5 | 6 | 9 | 14 |

b



10

a

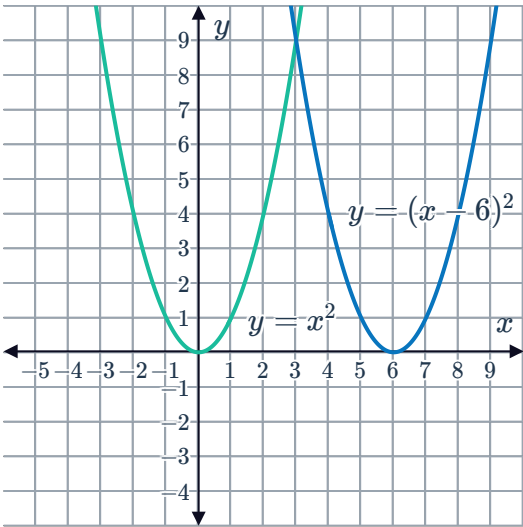
|     |   |   |   |   |   |
|-----|---|---|---|---|---|
| $x$ | 4 | 5 | 6 | 7 | 8 |
| $y$ | 4 | 1 | 0 | 1 | 4 |

b 0

c 6

e 6 units

f



11

a

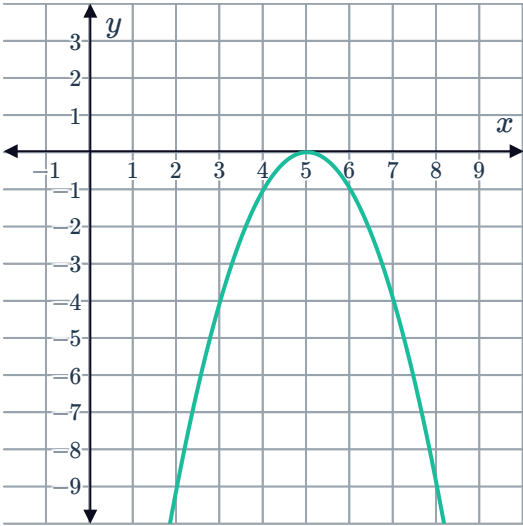
|     |    |    |   |    |    |
|-----|----|----|---|----|----|
| $x$ | 3  | 4  | 5 | 6  | 7  |
| $y$ | -4 | -1 | 0 | -1 | -4 |

b 0

c 5

d Vertex = (5, 0)

e



12

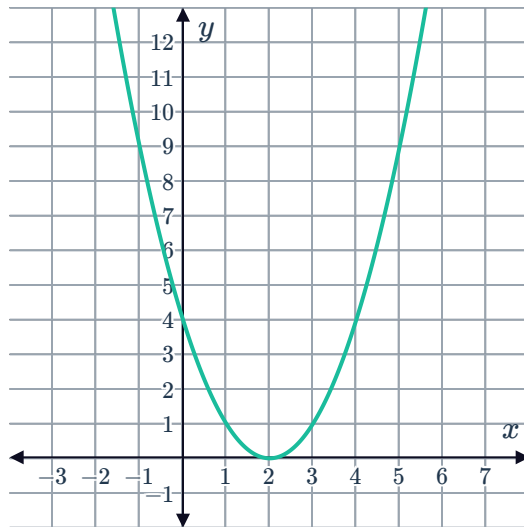
a

i  $x = 2$

ii  $y = 4$

iii Vertex  $(2, 0)$

iv

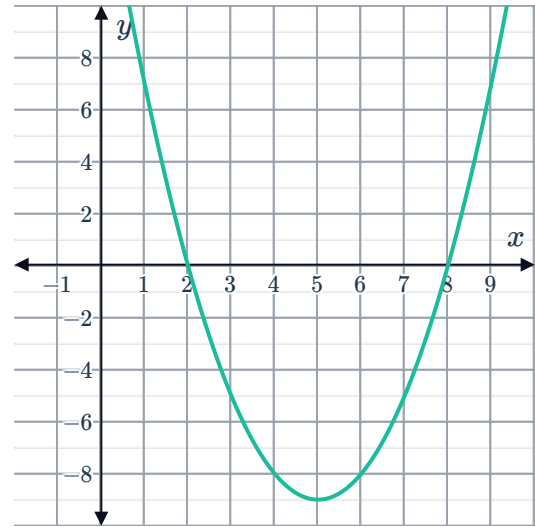


i  $x = 8, x = 2$

ii 16

iii Vertex  $(5, -9)$

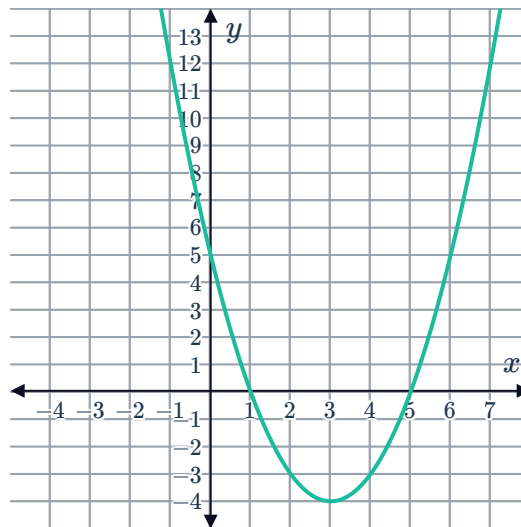
iv



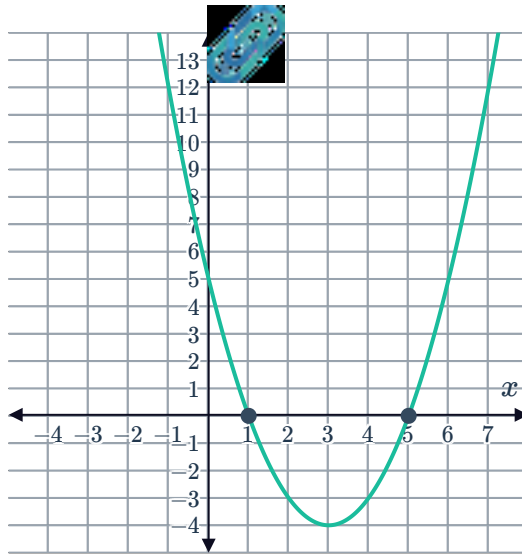
13

a  $x = 1, x = 5$

b



c

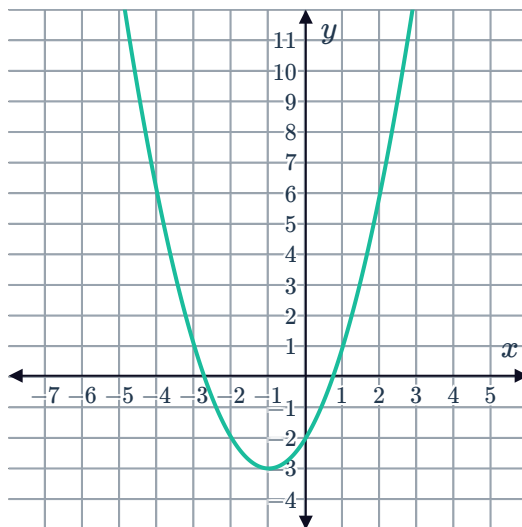


14

a  $x = -1$

b  $-3$

c



15

a  $x = -3$

b  $-4$

c  $(-3, -4)$

d  $(-3, -8)$

e

i Yes

ii Yes

iii No



16

- a  $-4$
- b  $y \geq -4$
- c All real  $x$

17

- a  $16$
- b  $y \leq 16$
- c All real  $x$

18

- a  $0$
- b  $y \geq 0$
- c  $x > 7$

19

- a  $0$
- b  $y \geq 0$
- c  $x < -4$

20

- a Minimum
- b  $-1$
- c  $y \geq -1$

21

- a Maximum
- b  $-2$
- c  $y \leq -2$

22

- a Minimum
- b  $-3$
- c  $y \geq -3$

??

a Maximum

b  $-3$

c  $y \leq -3$



24

a No

b  $x = 0$

c  $y = 0$